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Graph Retrieval Augmented Generation and Knowledge Graphs

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Abstract

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1 Introduction

In recent years, there has been a boom in artificial intelligence that is commercially available and accessible to the general public. When these large language models first emerged, the industry realized that the models were black boxes, and if they wanted the model to learn something new, they had to retrain it (an expensive process).

This caused three critical problems in LLMs:

- Hallucinations: The models confidently invent facts
- Knowledge gap: They cannot know anything that happened after their training
- Lack of privacy: They cannot access your private or corporate documents.

In September 2020, researchers at Facebook AI Research published the first paper on Retrieval-Augmented Generation (RAG). RAG was developed to address these issues; before the model responds, it searches a database for information related to the question asked and provides context for it.

These RAGs operate using semantic vector search: text is broken down into chunks, converted into numerical values, and stored. When you ask an LLM a question, the system searches for the chunks that are numerically most similar to your query, thereby providing the LLM with more context. This approach had a limitation: the LLM struggled to answer general questions based on the documents when using the RAG.

To address this issue, GraphRAG was developed—combining LLMs with knowledge graphs, which contain interconnected entities such as people, places, concepts, and relationships like "X works at Y." This allows us to extract broader contextual insights from the data at our disposal.

1.1 Context

1.2 Problem Statement

In this thesis, we aim to address the practical challenges involved in generating and using high-quality RAGs or GraphsRAGs. When using them, we identify three main issues: indexing time, indexing cost, and determining which systems are best suited to our needs—that is, the trade-offs involved in choosing one type of RAG over another.

1.3 Motivation and Objectives

2 Background and State of the Art

3 Proposed Solution

4 Evaluation

5 Discussions and Conclusions

6 Ethical, Deontological and Environmental Aspects

Bibliography

[1] Text example